Hodge Deligne Numbers for Hypersurfaces in Toric Varieties

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1 Mike Meeting 03 Aug

Mike took us through some of the basic ideas. It was a long meeting, but there were a lot of ideas, and so this is a writeup for some of the ideas from that meeting.

Any text in purple indicates either a question from myself or an incomplete thought.

1.1 Definition of Hodge-Deligne Numbers and Some Basic Properties

Let X be a variety over $\mathbb{C}$ of dimension[1] n. The *Hodge-Deligne numbers* denoted $e^{p,q}(X)$ for $0 \leq p, q \leq n$ are a collection of integers satisfying properties soon to be described. Moreover the polynomial $e_X(x, y) := \sum_{0\leq p,q\leq n} e^{p,q}(X)x^p y^q$ is called the *Hodge-Deligne* polynomial. The Hodge-Deligne numbers and polynomial satisfy the following properties

- The Hodge-Deligne polynomial is symmetric in x, y, $e_X(x, y) = e_X(y, x)$. In other words, the Hodge-Deligne numbers are symmetric in p, q, $e^{p,q}(X) = e^{q,p}$.
- If X is reducible then the Hodge-Deligne polynomial decomposes additively over the components: if $X = Y \amalg Z$ (presumably these have to be closed subsets?) then $e_X = e_Y + e_Z$. (A variety is by definition irreducible, so, perhaps the most general setting is for a separated, finite type over $\mathbb{C}$-scheme? Or maybe this property is supposed to be for closed subvarieties?)
- If X is a product variety then its Hodge-Deligne polynomial decomposes multiplicatively: if $X = Y \times_{\mathbb{C}} Z$ then $e_X = e_Y e_Z$.
- If X is projective and smooth (over $\mathbb{C}$) then recall that the hodge numbers are the dimensions of the $\mathbb{C}$-vector spaces given by cohomology over the sheaf of differentials

$$h^{p,q}(X) = \dim_{\mathbb{C}} H^q(X, \Omega^p_X).$$

[1] A variety is defined to be separated, irreducible, finite type k-scheme. I am trying to remember why we have that a variety has a well-defined dimension.

In this setting, the Hodge-Deligne numbers are related to the Hodge numbers in the following way

$$e^{p,q}(X) = (-1)^{p+q} \cdot h^{p,q}(X).$$

1.2 Examples

Example 1 (A point)**.** Let $X = \{*\} = \operatorname{Spec}(\mathbb{C})$. We have that X is smooth and projective. The following Hodge calculation will give that $e_X(x, y) = 1$.

Since X is affine $H^q(X, \Omega_X^p)$ is non-vanishing only for $q = 0$. Recall that $\Omega_X^p = 0$ for all $p > 0$ (I have forgotten why this is true, but it follows because X is a point. I think it should be easy to build the module of differentials in this case once one remembers the definition and construction) and so $h^{p,q} = 0$ for all $p, q > 0$. The only possible non-zero Hodge number we may have is for $p = q = 0$. We have the following

$$\begin{aligned}
h^{0,0}(X) &:= H^0(X, \Omega_X^0) \\
&= \dim_{\mathbb{C}} H^0(X, \mathcal{O}_X) && \text{By definition} \\
&= \dim_{\mathbb{C}} \Gamma(X, \mathcal{O}_X) && \text{By the cohomology axioms} \\
&= \dim_{\mathbb{C}} \mathbb{C} && X = \operatorname{Spec}(\mathbb{C}) \\
&= 1.
\end{aligned}$$

Overall, we have a that $h^{0,0}(X) = 1$ and $h^{p,q}(X) = 0$ for all other p, q. The claimed result follows.

Example 2 (Union of Points)**.** It follows from the previous example and the decomposition property of e_X that a disjoint union of n points has Hodge-Deligne polynomial $e_X(x, y) = n$.

Example 3 ($P^1_{\mathbb{C}}$)**.** Performing a similar Hodge computation to the example of the point one can compute that

$$e_{\mathbb{P}^1_{\mathbb{C}}} = 1 + xy.$$

This might eb a worthwhile computation. $\mathbb{P}^1$ is covered by two affine patches and so we only need to compute for $p < 2$. We need to recall how to compute the sheaf of differentials however, but I think it should be nonvanishing only degree $0, 1$.

Example 4 ($\mathbb{A}^1_{\mathbb{C}}$)**.** We can decompose $P^1 = \mathbb{A}^1 \amalg \{*\}$ the affine line with a "point at infinity". "point at infinity." Since $\mathbb{A}^1$ is an open subset and $\{*\}$ is a closed point we have

$$e_{\mathbb{A}^1_{\mathbb{C}}}(x, y) = e_{\mathbb{P}^1_{\mathbb{C}}} - e_{\{*\}} = xy.$$

~~I'm a bit confused by this example, becuase the affine line is affine, and so we should expect it to only~~

~~have have cohomology in degree 0. That is to say, I would expect that this should be a polynomial only in y.~~ The Hodge-Deligne numbers agree with the Hodge numbers only when X is projective – $\mathbb{A}^1$ is not projective, so we're gucci.

Example 5 ($\mathbb{A}^n_{\mathbb{C}}$.)**.** Using the product property of the Hodge-Deligne polynomial we have

$$E_{\mathbb{A}^n} = (xy)^n.$$

From this example, we can read off the hodge numbers without needing to do any Hodge comptuations.

Example 6 (Algebraic Torii)**.** We can decompose the affine line $\mathbb{A}^1 = \mathbb{C}^* \amalg \{*\}$ and so using the decomposition property of the Hodge-Deligne polynomial we have

$$e_{\mathbb{C}^*} = e_{\mathbb{A}^1_{\mathbb{C}}} + e_{\{*\}} = xy - 1.$$

Then, using the product property we further have that

$$e_{(\mathbb{C}^*)^n} = (e_{\mathbb{C}^*})^n = (xy-1)^n.$$

Example 7 ($\mathbb{P}^n_{\mathbb{C}}$)**.** Recall that we can decompose $\mathbb{P}^n = \mathbb{C}^n \amalg \mathbb{P}^{n-1}$ and then we can find the Hodge-Deligne polynomial of $\mathbb{P}^n$ inductively. Consider,

$$\begin{aligned}
e_{\mathbb{P}^1} &= 1 + xy \\
e_{\mathbb{P}^2} &= e_{\mathbb{C}^2} + e_{\mathbb{P}^1} = (xy)^2 + (1 + xy) \\
e_{\mathbb{P}^3} &= e_{\mathbb{C}^3} + e_{\mathbb{P}^2} = (xy)^3 + (1 + xy + (xy)^2) \\
&\dots \\
e_{\mathbb{P}^n} &= 1 + (xy) + (xy)^2 + \cdots + (xy)^n.
\end{aligned}$$

2 Procedure for Hypersurfaces in Toric Varities.

we should go back and fill out some of the preamble thoery here

Suppose we have a full dimension polytope Δ. We can decompose our toric variety

$$P_\Delta = \amalg_{\Gamma' \leq \Delta} T_{\Gamma'}$$

where $T_{\Gamma'} = (\mathbb{C}^*)^{\dim \Gamma'}$. And so we can compute the Hodge-Deligne polynomial of a toric variety using the decomposition property. Does each subcone really just give a torus in a toric variety? Or is it somehow sufficient to compute the Hodge-Deligne polynomial on a dense open?

Let $Z_{\Gamma'} = Z \cap T_{\Gamma'}$. We have the following formula

$$e^{pq}(\bar{Z}) = e^{pq}(Z) + \sum_{\Gamma' < \Delta} e^{pq}(Z_{\Gamma'}).$$

Note that a consequence of this formula is that for $p = n$ or $q = n$, $e^{pq}(\bar{Z})$ is determined by $e^{pq}(Z)$. I think this should follow from the dimension property of Hodge-Deligne numberes, and each $e^{pq}(Z_{\Gamma'})$ has dimension strictly less than n. Is this reasoning correct?

We also have the following formulae

$$e^{pq}(Z) = e^{p+1,q+1}(T) = \begin{cases} 0 & p \neq q \\ (-1)^{n+p+1}\binom{n}{p+1} & p = q \end{cases} \tag{1}$$

for $p + q \geq n$. And,

$$\sum_{q=0}^{n-1} e^{pq}(Z) = (-1)^p \left((-1)^{n+1}\binom{n}{p+1} - \sum_{k \geq 1} (-1)^k \binom{n+1}{p+k+1} \ell^*(k\Delta) \right). \tag{*}$$

where ℓ^* denotes the number of interior lattice points of the given polytope. Is this last formula actually correct? Are there conditions on p, q for when this formula holds?

I think the algorithm is then as follows: we iteratively compute the Hodge-Deligne numbers starting from the smalles cone, and we only need to use formula $(*)$ to add the numbers in degree n, is that correct?

2.1 Examples

Example 8 (Hypersurface in $\mathbb{P}^1 \times \mathbb{P}^1$)**.** Recall that the projective toric variety $\mathbb{P}^1 \times \mathbb{P}^1$ is given by the polytope $\Delta = \text{ConvexHull}((1,1),(1,-1),(-1,1),(-1,1))$, the reflexive square with the origin as the interior lattice point. We use the recursive procedure described above to compute the Hodge-Deligne numbers for the hypersurface defined by

$$f(x,y) = xy + \frac{x}{y} + \frac{1}{xy} + \frac{y}{x}.$$

Note that the Newton polytope of this Laurent polynomials is in fact Δ. First we determine that this polynomial is non-degenerate with respect to Δ. Let us label each of the facets (edges) of Δ as follows; $\Gamma_1 = \text{ConvexHull}((-1,1),(1,1))$ as the "top face", $\Gamma_2 = \text{ConvexHull}((1,1),(1,-1))$ as the "right face", and proceeding clockwise. The facet[2] polynomials are then as follows

$$\begin{aligned} f_{\Gamma_1} &= \frac{y}{x} + xy \\ f_{\Gamma_2} &= xy + \frac{x}{y} \\ f_{\Gamma_3} &= \frac{x}{y} + \frac{1}{xy} \\ f_{\Gamma_4} &= \frac{1}{xy} + \frac{y}{x}. \end{aligned}$$

We show that $Z(f_{\Gamma_1})$ is smooth. The partial derivatives are

$$\frac{\partial f_{\Gamma_1}}{\partial x} = -\frac{y}{x^2} + y \qquad \frac{\partial f_{\Gamma_1}}{\partial y} = \frac{1}{x} + x.$$

$\partial f_{\Gamma_1}/\partial y = 0$ has solutions in $(\mathbb{C}^*)^2$ with x-coordinate $x = \pm 1$ whereas $\partial f_{\Gamma_1}/\partial x = 0$ implies $x = \pm i$. This system has no simultaneous solutions in $(\mathbb{C}^*)^2$ and so $Z(f_{\Gamma_1})$ is smooth. A similar calculation will give that each $Z(f_{\Gamma_i})$ is smooth and thus f is indeed non-degenerate with respect to Δ.

Next we compute the Hodge-Deligne numbers for $Z = Z(f)$. Since we are in $n = 2$, in principle, we have nonvanishing Hodge-Deligne numbers for $p, q \leq n$; However formula (1) above implies that for $p + q \geq 2$ the only nonvanishing Hodge number is $e^{11}(Z) = (-1)^{2+1+1}\binom{2}{1+1} = 1$.

Next we compute the contributions from $\sum_{\Gamma' < \Delta} e^{pq}(Z_{\Gamma'})$. Let us first consider $\Gamma' = \Gamma_1$. By definition this is $Z(f_{\Gamma_1}) \cap T_{\Gamma_1} \cong Z(f_{\Gamma_1}) \cap \mathbb{C}^*$. Recall $f_{\Gamma_1} = y/x + xy = y(1/x + x)$. The primitive normal vector corresponding to Γ_1 is $(0,1)$, and so intersecting with T_{Γ_1} algebraically corresponds to sending $y \mapsto 1$ (I'm not this part is articulated correctly; I think restricting to this cone corresponds to restricting to an open patch where we "divide by y", similar to the distinguished patches in projective space? Is that correct?). And so we search for solution in $\mathbb{C}^*$ to $1/x + x = 0$. The solution set is $x = \pm i$; topologically this is the disjoint union of two points in $\mathbb{C}^*$. It follows then from the union-sum property of Hodge-Deligne polynomials and from the Hodge-Deligne polynomial of a point found above that $e_{Z_{\Gamma_1}} = 2$. A very similar argument yields that $e_{Z_{\Gamma'}} = 2$ for all $\Gamma < \Delta$. And so $e^{00}(\bar{Z}) = 8$.

The last Hodge-Deligne numebr to compute is $e^{10}(Z)$, we compute this using the formula $(*)$. We have

[2] I think in the general setting the vertex monomials are always vacuously smooth since I think they do not even have solutions in $(\mathbb{C}^*)^2$. This argument about the general setting should be checked, but one can certainly verify here that the vertex monomials are smooth.

$$
\begin{aligned}
e^{10}(Z) + e^{11}(Z) &= (-1)^1 \left[(-1)^{2+1} \binom{2}{1+1} - \sum_{k \geq 1} (-1)^k \binom{2+1}{1+k+1} \ell^*(k\Delta) \right] \\
e^{10}(Z) + 1 &= - \left[(-1) - (-1) \binom{3}{3} (1) \right] \\
e^{10}(Z) &= -1.
\end{aligned}
$$

By p, q symmetry of the Hodge-Deligne numbers, we[3] also have $e^{01}(Z) = -1$.

Overall, we have found the Hodge-Deligne polynomial for f as

$$e_Z(x, y) = 8 - x - y + xy.$$

Is there a good way to check if this makes sense?

- TODO
- I think we also need to compute the e^{00} contributions from Z, perhaps we need to do this with the lattice point counting formula?
- Write up computation of a hypersurface on a d8

[3] As an aside, interestingly, this portion of the calculation seems to not have depended on the choice of hypersurface at all, is this correct?

3 graveyard

Example 9 ($\mathbb{P}^1 \times \mathbb{P}^1$)**.** Here we compute the Hodge-Deligne numbers for $Z = \mathbb{P}^1 \times \mathbb{P}^1$. Recall that for this toric variety we have $\Delta = \text{ConvexHull}((1,1),(1,-1),(-1,1),(-1,1))$. Since $n = 2$ we have that there are only four Hodge-Deligne numbers to compute.

We start with $p = 1, q = 1$. This is the only case for which $p + q \geq 2$ and we have

$$e^{11}(Z) = e^{2,2}(T) = (-1)^{2+1+1}\binom{2}{1+1} = 1.$$

Next we compute e^{01}. For this we use $(*)$. We have

$$\begin{aligned} e^{10} + e^{11} &= (-1)^1 \left[(-1)^{2+1}\binom{2}{1+1} - \sum_{k\geq 1} (-1)^k \binom{1+1}{1+k+1} \ell^*(k\Delta) \right] \\ e^{10} + 1 &= (-1)\left[(-1) - \left((-1)^1 \binom{3}{3} \cdot 1 \right) \right] \\ e^{10} &= -1. \end{aligned}$$

Then by p, q symmetry we have that $e^{01} = -1$ also.

Lastly, we compute e^{00}. We can do this by recurring and computing the Hodge-Deligne number for a unit ray. Recall that the toric variety corresponding to a single ray is the affine line $\mathbb{C}^1$. We computed earlier that this has Hodge-Deligne polynomial $e_{\mathbb{C}^1} = xy$. Meaning that the Hodge numbers for $\mathbb{C}^1$ are all zero except for $p = 1, q = 1$.